

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.5	10.2	0.1	0	0	66.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25	8.9	0.1	0	0	66
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	24.1	9.9	0.1	0	0	65.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.6	8.2	0.1	0	0	66.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	24.5	8.9	0.3	0	0	66.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.9	7.9	0.1	0	0	66.1
U2 - Efficacy additive alpha prior Beta(1, 1)	24.4	9.2	0.1	0	0	66.3
U3 - Efficacy additive alpha prior Beta(1, 1)	25.3	8	0.1	0	0	66.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.9	3.1	1	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17	3	1	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.9	3.1	1	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.9	3.1	1	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17	3.1	1	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17	3	1	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	17	3	1	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	17	3	1	0.2	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.8	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.7	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.8	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.8	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	1.8	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	1.8	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.8	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	1.8	0.3	0.1	0	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	37.6	26.7	27.2	6	0.5	2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	40.2	27.4	24.2	5.7	0.5	2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	37.6	26	28.1	5.6	0.6	2.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	41.2	25.6	25.2	5.6	0.4	2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	38.6	23.9	28.4	6.4	0.7	2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	43.6	22.9	25.7	5.2	0.6	2
U2 - Efficacy additive alpha prior Beta(1, 1)	38.8	23.3	29.3	5.9	0.7	2
U3 - Efficacy additive alpha prior Beta(1, 1)	43.5	22.6	26.1	5.2	0.6	2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.2	6.5	6.4	2.8	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.2	6.5	6.3	2.8	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.2	6.4	6.5	2.7	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.2	6.3	6.4	2.8	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.3	6.3	6.5	2.7	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.3	6.2	6.6	2.7	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	13.3	6.2	6.6	2.7	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	13.3	6.3	6.5	2.7	0.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.7	1.2	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.7	1.2	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.6	1.2	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.6	1.2	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.6	1.2	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.6	1.2	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.6	1.2	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.6	1.2	0.6	0.2	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.6	47.4	17.5	5.3	0.4	10.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.5	49.6	14.2	4.8	0.2	10.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.9	47.7	17.9	4.4	0.4	10.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.5	49.2	15	4.2	0.4	10.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.3	47	17.6	4.7	0.6	10.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.5	47.9	15	4.5	0.3	10.8
U2 - Efficacy additive alpha prior Beta(1, 1)	19.9	46.8	17.4	4.5	0.6	10.8
U3 - Efficacy additive alpha prior Beta(1, 1)	21.8	48.3	14.7	4.1	0.3	10.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.4	8.6	5.3	2.6	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.4	8.6	5.3	2.6	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.4	8.5	5.4	2.6	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.4	8.6	5.4	2.6	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.4	8.5	5.5	2.5	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.4	8.4	5.5	2.5	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.4	8.5	5.4	2.6	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.4	8.4	5.4	2.6	0.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.1	1	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.1	1	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.1	1.1	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.1	1	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.1	1	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.1	1	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.1	1	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.1	1	0.7	0.2	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.8	11.8	24.4	42.7	10.3	2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.7	12.9	21.4	42.4	8.6	2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	10.6	24	44.1	10.1	2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14	11.8	20.6	42.8	8.8	2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.5	12.2	23.8	43.1	10.4	2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.5	12.3	21.3	42.4	8.5	2
U2 - Efficacy additive alpha prior Beta(1, 1)	8.5	11.2	25.3	42.3	10.7	2
U3 - Efficacy additive alpha prior Beta(1, 1)	13.8	11.7	21.1	42.6	8.8	2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.9	5.1	7.2	8.1	4.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.9	5.1	7.2	8.2	4.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.9	5.1	7.3	8.1	4.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.9	5.1	7.3	8.1	4.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.9	5	7.4	8	4.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.9	5	7.3	8.1	4.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.9	5	7.4	8.1	4.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.9	5	7.3	8.1	4.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3	1.5	2.2	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.1	1.5	2.2	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.1	1.5	2.2	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.1	1.5	2.2	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.1	1.5	2.2	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.1	1.5	2.2	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.1	1.5	2.2	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.1	1.5	2.2	0.6	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5	37.9	24.4	21.8	6.7	4.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8	42.5	23.6	17.6	4.1	4.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.1	35.3	26.6	21.5	7.3	4.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.2	41.6	24.4	17.4	4.2	4.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.6	33.8	27.6	21.3	7.5	4.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8	39.3	26.3	17.9	4.3	4.2
U2 - Efficacy additive alpha prior Beta(1, 1)	6	34.1	27.1	21.2	7.4	4.2
U3 - Efficacy additive alpha prior Beta(1, 1)	8.1	39.4	25.7	18.1	4.5	4.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.2	7.2	6.7	6.1	4.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.3	7.1	6.7	6.1	4.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.3	7	6.8	6.1	4.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.3	7	6.8	6	4.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.3	6.9	6.8	6	4.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.3	6.9	6.9	6	4.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.3	6.9	6.8	6.1	4.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.3	6.9	6.8	6.1	4.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.4	1.6	2.1	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.4	1.6	2.2	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.4	1.6	2.1	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.4	1.6	2.2	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.3	1.6	2.1	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.3	1.6	2.2	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.3	1.6	2.2	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.3	1.6	2.2	0.9	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	70.3	12.7	1.8	0.2	0	15
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	72.1	11.4	1.4	0.1	0	15
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	71.6	11.6	1.8	0.2	0	14.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	72.7	11	1.1	0.2	0	15
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	72.7	10.1	1.6	0.3	0	15.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	73.7	9.6	1.3	0.2	0	15.2
U2 - Efficacy additive alpha prior Beta(1, 1)	73.1	10.2	1.6	0.2	0	14.9
U3 - Efficacy additive alpha prior Beta(1, 1)	74.4	9.2	1.3	0.2	0	14.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.9	3.7	1.4	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.9	3.7	1.4	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	22	3.6	1.4	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.9	3.6	1.4	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	22	3.5	1.4	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22	3.5	1.4	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	22	3.5	1.4	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	22	3.5	1.4	0.3	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.5	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.5	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.5	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.5	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.5	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.5	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.5	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.5	0.3	0.1	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.9	45.9	15.5	7.4	0.8	2.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.2	48.8	12.6	5.4	0.5	2.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.1	44.8	16.4	7	1.2	2.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.1	47.5	13.4	6	0.5	2.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	29.1	43.2	16.4	7.8	1	2.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	30.3	46.6	13.8	6.1	0.7	2.5
U2 - Efficacy additive alpha prior Beta(1, 1)	28.9	44.2	16.4	7.2	0.8	2.5
U3 - Efficacy additive alpha prior Beta(1, 1)	30.4	47.1	13.6	5.7	0.7	2.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.4	8.1	5	2.9	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.4	8.2	5	2.9	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.5	8	5.1	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.5	8	5.1	2.9	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.5	7.9	5.2	2.9	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.5	7.9	5.2	2.8	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	12.5	7.9	5.2	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.5	7.9	5.1	2.9	1.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	0.9	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	0.9	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	0.9	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	0.9	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	0.9	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	0.9	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	0.9	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	0.9	0.8	0.3	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	42.6	6.6	0.6	0.2	0	50
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	43	6.4	0.4	0.2	0	50
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	42.4	6.6	0.8	0.2	0	50
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	43.2	6.3	0.6	0.2	0	49.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	42.8	6.7	0.6	0.2	0	49.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	43.2	6.3	0.5	0.2	0	49.8
U2 - Efficacy additive alpha prior Beta(1, 1)	42.5	6.4	0.8	0.2	0.1	50
U3 - Efficacy additive alpha prior Beta(1, 1)	43.3	6.2	0.5	0.2	0	49.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.5	2.7	0.9	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.4	2.7	0.9	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.4	2.7	0.9	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.5	2.7	0.9	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.5	2.6	0.9	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.5	2.6	0.9	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	19.5	2.6	0.9	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	19.5	2.6	0.9	0.1	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	1.4	0.2	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	1.4	0.2	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	1.4	0.2	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	1.4	0.2	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	1.4	0.2	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	1.4	0.2	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	1.4	0.2	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	1.4	0.2	0.1	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.9	19.5	17.6	11.3	11.4	20.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.4	19.4	15.1	10.1	9.8	20.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.1	19.5	18.8	10.9	11.3	20.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.4	18.8	15.5	10.2	9.7	20.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.8	19.3	17.7	10.9	11.9	20.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.1	18.6	15.1	10.4	9.5	20.3
U2 - Efficacy additive alpha prior Beta(1, 1)	19.4	20.1	17.9	10.8	11.6	20.2
U3 - Efficacy additive alpha prior Beta(1, 1)	26.2	19.2	15.1	10	9.2	20.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.6	5.6	5.2	3.3	2.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.4	5.8	5.2	3.4	2.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.4	5.7	5.3	3.4	2.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.5	5.6	5.3	3.4	2.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.5	5.6	5.3	3.3	2.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.5	5.5	5.4	3.3	2.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.6	5.5	5.3	3.3	2.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.5	5.5	5.3	3.4	2.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3	1.3	1.2	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3	1.2	1.2	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3	1.3	1.1	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	2.9	1.2	1.2	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3	1.3	1.2	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	2.9	1.3	1.2	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3	1.3	1.2	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	2.9	1.3	1.2	0.4	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.5	31.8	7.4	1.4	0.4	44.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.7	32.8	6.3	1.1	0.4	44.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.7	32.5	7.1	1.2	0.7	44.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.5	33	6.2	1.2	0.4	44.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.1	31.7	7.6	0.9	0.9	44.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.3	31.8	6.9	0.8	0.6	44.6
U2 - Efficacy additive alpha prior Beta(1, 1)	13.9	32.1	7.6	1	0.8	44.6
U3 - Efficacy additive alpha prior Beta(1, 1)	14.8	31.9	7.4	0.8	0.6	44.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.1	6	2.8	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.1	6	2.8	1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.1	5.9	2.9	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.1	5.9	2.9	1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14	5.9	2.9	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.1	5.9	2.9	1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	14.1	5.9	2.9	1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	14.1	5.9	2.9	0.9	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.3	0.6	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.3	0.6	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.3	0.6	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.3	0.6	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.3	0.6	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.3	0.6	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.3	0.6	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.3	0.6	0.3	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	66.3	24.3	1.4	0	0	8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	67.7	23.1	1	0	0	8.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	67.2	23.3	1.5	0	0	8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	69	22	1.1	0	0	7.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	68.6	22.1	1.1	0.1	0	8.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	71.5	19.9	0.6	0	0	8
U2 - Efficacy additive alpha prior Beta(1, 1)	68.5	22.5	0.7	0.1	0	8.2
U3 - Efficacy additive alpha prior Beta(1, 1)	71.2	19.9	0.8	0	0	8.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.9	5.7	1.9	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.9	5.7	1.9	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21	5.6	1.9	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.1	5.5	1.9	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.1	5.4	1.9	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.2	5.4	1.9	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	21.2	5.4	2	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	21.2	5.4	1.9	0.3	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	2.3	0.5	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	2.3	0.5	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	2.3	0.5	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	2.3	0.5	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	2.3	0.5	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	2.3	0.5	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.3	0.5	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	2.3	0.5	0.2	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.5	18.1	13.4	35.3	13.4	3.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.8	18.1	8.7	35.4	11.7	3.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15	18.4	12.1	38.3	12.9	3.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.6	16.8	8.6	35.5	12.2	3.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.6	18.1	13.6	37.4	13	3.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.6	17	9.7	35.5	11.9	3.3
U2 - Efficacy additive alpha prior Beta(1, 1)	14.3	18.6	12.5	37.9	13.4	3.3
U3 - Efficacy additive alpha prior Beta(1, 1)	22	16.4	9.8	36.2	12.3	3.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	5.7	6.1	7.6	4.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	5.7	6.2	7.4	4.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.4	5.7	6.2	7.5	4.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.4	5.6	6.2	7.5	4.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.4	5.6	6.2	7.5	4.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.5	5.6	6.1	7.6	4.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.5	5.7	6.1	7.6	4.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.5	5.5	6.1	7.6	4.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.2	1.5	2.3	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.2	1.4	2.2	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.2	1.5	2.3	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.2	1.5	2.3	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.2	1.5	2.3	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.2	1.5	2.3	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.2	1.5	2.2	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.2	1.5	2.3	0.7	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.7	33.6	8.4	1.9	0.2	39.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.1	33.6	6.9	1.7	0.2	39.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.1	34	7.7	1.9	0.2	39.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.2	33.1	8.1	1.6	0.2	39.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16	34.2	8.4	1.8	0.3	39.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.8	32.9	8	1.5	0.3	39.5
U2 - Efficacy additive alpha prior Beta(1, 1)	15.8	34	8.2	2.1	0.3	39.6
U3 - Efficacy additive alpha prior Beta(1, 1)	17.8	33.8	7.2	1.8	0.2	39.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.8	6.2	3.2	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.8	6.3	3.3	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.8	6.3	3.3	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.8	6.2	3.3	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.8	6.3	3.2	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.8	6.2	3.3	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	13.8	6.2	3.3	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	13.8	6.2	3.3	1.1	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.5	0.7	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.5	0.7	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	0.7	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	0.7	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	0.7	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.5	0.7	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.5	0.7	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.5	0.7	0.4	0.1	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.3	25.6	14.1	23.1	7.3	7.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.9	22.9	10.4	20.6	5.5	7.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.5	25.1	15.1	21.9	7.8	7.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	34.1	21.6	10.4	20.1	6.1	7.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.4	24.3	14.8	22	7.9	7.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.3	21.8	9.8	19.5	6.9	7.7
U2 - Efficacy additive alpha prior Beta(1, 1)	23.1	24	15.8	21.5	7.9	7.7
U3 - Efficacy additive alpha prior Beta(1, 1)	33	23.1	10.2	19.5	6.5	7.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.8	6.6	6.3	5.3	2.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.8	6.6	6.2	5.3	2.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.8	6.6	6.3	5.2	3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.8	6.6	6.3	5.2	3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.8	6.5	6.3	5.2	3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.8	6.5	6.4	5.1	3.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.9	6.5	6.3	5.1	3.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	7.8	6.5	6.2	5.3	3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.5	1.5	1.8	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.5	1.5	1.8	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.5	1.5	1.8	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.5	1.5	1.7	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.5	1.5	1.8	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.6	1.5	1.8	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.5	1.5	1.8	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.5	1.5	1.8	0.5	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	48.6	34.3	7.5	1.8	0.2	7.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	56.9	29.3	4.8	1.1	0.1	7.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	48.9	34.1	7.6	1.6	0.2	7.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	58.2	28.7	4.4	1.1	0.1	7.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	52	30.7	7.3	2.1	0.3	7.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	61.6	25.2	4.2	1.2	0.1	7.7
U2 - Efficacy additive alpha prior Beta(1, 1)	51	31.7	7.6	2	0.2	7.5
U3 - Efficacy additive alpha prior Beta(1, 1)	60.6	26.5	3.9	1.2	0.2	7.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.1	8.2	5	2.1	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.1	8.1	5.1	2	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.2	8	5.1	2.1	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.1	8.1	5	2.1	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.3	8	5	2.1	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.3	8	5	2.1	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	13.3	8	5	2.1	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	13.3	7.9	5	2.1	0.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.7	1.3	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.7	1.3	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.7	1.3	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.7	1.3	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	3.8	1.3	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	3.8	1.3	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	3.8	1.3	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.2	3.8	1.3	0.9	0.2	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.8	31.3	15.9	17.4	7.5	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	33.5	35.9	13.5	13.2	3.8	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.6	30.1	16.2	17.2	8.8	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	35.4	34.4	13.4	12.3	4.4	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.8	26	16.2	17.9	8	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	37.4	32.1	14.9	11.3	4.2	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	31	26.6	16.5	18	7.8	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	37.5	32.7	14.1	11.2	4.4	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.6	6.2	5.5	6.2	6.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.6	6.2	5.6	6.1	6.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.7	6	5.5	6.1	6.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.7	6	5.6	6.1	6.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.8	5.9	5.5	6	6.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.8	5.9	5.5	6	6.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.8	5.9	5.5	6.1	6.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.8	5.9	5.5	6	6.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	3.3	1.8	2.7	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	3.4	1.8	2.7	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.3	1.7	2.7	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	3.3	1.8	2.7	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.3	1.8	2.6	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	3.2	1.8	2.7	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	3.2	1.8	2.7	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	3.2	1.8	2.7	1.4	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	29.9	58.1	11.1	0.6	0	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.5	58.1	8.7	0.4	0	0.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.2	56.9	12	0.6	0	0.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	33.3	56.8	9.1	0.5	0	0.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.6	54.9	12.7	0.5	0	0.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.4	55.1	9.8	0.4	0	0.3
U2 - Efficacy additive alpha prior Beta(1, 1)	31.8	54.7	12.7	0.5	0	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	34.3	55.1	9.9	0.4	0	0.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.4	11.1	5.9	1.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.4	11.1	5.9	1.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.5	10.9	6	1.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.5	10.9	6	1.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.6	10.7	6.1	1.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.7	10.7	6.1	1.3	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.7	10.7	6.1	1.3	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.7	10.7	6.1	1.3	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3	1	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3	1	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3	1	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3	1	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3	1	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3	1	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3	1	0.4	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3	1	0.4	0	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	14.4	11.3	19.8	44.5	0.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.6	15.8	11.2	19.2	38.5	0.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.4	13.3	11.8	20.4	45.4	0.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.8	16	11.1	18.5	38.9	0.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7	14.1	13.1	19.1	46	0.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.7	15.7	11.3	18.6	40	0.7
U2 - Efficacy additive alpha prior Beta(1, 1)	7.9	12.5	13.7	18.9	46.3	0.7
U3 - Efficacy additive alpha prior Beta(1, 1)	14.5	15.5	12.3	17.9	39.1	0.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.8	4.4	5.1	6.3	10.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.8	4.3	5.1	6.3	10.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.8	4.2	5.1	6.3	10.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.8	4.3	5.1	6.2	10.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	4.2	5.1	6.2	10.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	4.2	5.1	6.2	10.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.8	4.2	5.2	6.1	10.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.9	4.2	5.1	6.2	10.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.9	1.5	2.5	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.9	1.5	2.5	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.5	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.9	1.5	2.4	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.5	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.4	2.4	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.9	1.5	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.9	1.5	2.4	1.4	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	23	56.1	11.3	0.4	0	9.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.3	55.9	9.5	0.2	0	9.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.4	54.9	12.1	0.4	0	9.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.8	54.2	9.4	0.5	0	9.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.7	56.2	12.4	0.6	0	9.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.7	54	10.6	0.5	0	9.2
U2 - Efficacy additive alpha prior Beta(1, 1)	22.3	55.6	12.5	0.5	0	9.1
U3 - Efficacy additive alpha prior Beta(1, 1)	25.4	54.6	10.1	0.5	0	9.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.1	11	5.4	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.1	11.1	5.3	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.1	11	5.4	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.1	10.9	5.4	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.1	10.9	5.5	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.2	10.8	5.5	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.2	10.9	5.4	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.2	10.9	5.4	1.1	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.7	1.1	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.7	1	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.7	1	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.7	1	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.7	1	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.7	1	0.5	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.7	1	0.5	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.7	1	0.5	0	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.1	52.2	24.8	6	0.5	4.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.4	52.5	22.3	5.4	0.1	4.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.8	51.7	25.9	5.8	0.5	4.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.9	52.9	21.9	4.8	0.2	4.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11	51.4	26.6	6.3	0.4	4.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.6	52.5	21.9	4.5	0.2	4.3
U2 - Efficacy additive alpha prior Beta(1, 1)	11.6	50.8	27	5.9	0.4	4.3
U3 - Efficacy additive alpha prior Beta(1, 1)	17.4	52.1	21.4	4.6	0.2	4.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.4	10.2	8.2	3.4	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.4	10.3	8.2	3.3	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.4	10.1	8.3	3.4	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.4	10.2	8.3	3.3	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.4	10.1	8.4	3.4	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.4	9.9	8.4	3.5	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.5	10	8.4	3.4	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	6.5	9.9	8.4	3.4	1.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.8	1.7	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.8	1.7	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.9	1.7	1.3	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.9	1.7	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.8	1.7	1.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.8	1.7	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.8	1.7	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.8	1.7	1.2	0.3	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	57.1	1.7	0	0	0	41.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	57	1.5	0	0	0	41.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	57.3	1.6	0	0	0	41.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	57	1.5	0	0	0	41.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	57.2	1.5	0	0	0	41.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	56.9	1.5	0	0	0	41.6
U2 - Efficacy additive alpha prior Beta(1, 1)	57.1	1.6	0	0	0	41.3
U3 - Efficacy additive alpha prior Beta(1, 1)	57	1.5	0	0	0	41.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.4	1.5	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.4	1.5	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.4	1.5	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.4	1.5	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.4	1.5	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.4	1.5	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	21.4	1.5	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	21.4	1.5	0.2	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	0.6	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	0.6	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	0.6	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	0.6	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	0.6	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	0.6	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	0.6	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	0.6	0.1	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.3	59.8	9.9	0.4	0	4.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	29.5	57.4	8	0.3	0	4.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.1	59.4	9.7	0.3	0	4.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.8	56.2	8	0.3	0	4.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.5	58.4	10.2	0.3	0	4.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	30.3	56.2	8.3	0.3	0	4.9
U2 - Efficacy additive alpha prior Beta(1, 1)	26.6	58.5	9.9	0.4	0	4.6
U3 - Efficacy additive alpha prior Beta(1, 1)	30.3	56.9	7.8	0.4	0	4.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.8	12.7	4.9	0.8	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.7	12.7	4.9	0.8	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.8	12.6	5	0.8	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.8	12.7	4.9	0.8	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.9	12.5	5	0.8	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.9	12.5	5	0.8	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.9	12.4	5.1	0.8	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.9	12.5	5	0.8	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.1	1	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4	1	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.1	1	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.1	1	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.1	1	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.2	1	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.1	1	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.2	1	0.3	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.1	5.8	0	0	0	79.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.9	5.6	0	0	0	78.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.3	6.2	0	0	0	78.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.7	5.5	0	0	0	78.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.3	6.3	0	0	0	78.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.9	5.2	0	0	0	78.9
U2 - Efficacy additive alpha prior Beta(1, 1)	15.1	5.6	0	0	0	79.3
U3 - Efficacy additive alpha prior Beta(1, 1)	15.9	5.3	0	0	0	78.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.6	1.7	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.6	1.7	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.6	1.7	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.6	1.7	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.6	1.7	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.6	1.7	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	17.5	1.7	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	17.6	1.7	0.2	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	1.3	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	1.2	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	1.2	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	1.2	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	1.2	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	1.2	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	1.3	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	1.3	0.1	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	1.8	5.5	53.6	37.9	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.7	2.9	7	55.6	32.8	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	1.3	5.3	52.7	39.6	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.9	2.6	6.6	54.8	34.1	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	1	4.8	50.9	42.3	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	2	1.5	6.3	52.9	37.3	0
U2 - Efficacy additive alpha prior Beta(1, 1)	1	1.1	4.5	50.7	42.7	0
U3 - Efficacy additive alpha prior Beta(1, 1)	1.7	2.1	6.2	52.9	37.1	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.2	3.2	3.7	9.3	10.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.2	3.2	3.7	9.3	10.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.2	3.2	3.7	9.3	10.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.2	3.2	3.7	9.2	10.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.2	3.1	3.7	9.2	10.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.2	3.1	3.7	9.2	10.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.2	3.1	3.7	9.1	10.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.2	3.1	3.7	9.1	10.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.5	1.1	3.2	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.5	1.2	3.2	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.2	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0	2.5	1.2	3.2	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.5	1.1	3.2	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.5	1.1	3.2	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0	2.5	1.1	3.2	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0	2.5	1.1	3.2	1.2	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	33.5	35.7	27.7	2.4	0	0.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	38.5	36.1	22.3	2.4	0	0.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.8	36.2	28	2.3	0	0.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	37.4	36.6	22.7	2.6	0	0.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	33.6	35.4	27.5	2.8	0	0.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	39.4	34.5	22.7	2.7	0	0.7
U2 - Efficacy additive alpha prior Beta(1, 1)	33.9	35.3	27.4	2.7	0	0.7
U3 - Efficacy additive alpha prior Beta(1, 1)	38.9	34.8	23.4	2.2	0	0.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.5	9.8	7.7	1.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.5	9.7	7.8	1.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.5	9.6	7.9	1.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.5	9.5	8	1.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.7	9.3	8	1.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.7	9.3	8	1.7	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.6	9.4	7.9	1.7	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.7	9.3	8	1.7	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	3.3	1.4	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	3.4	1.4	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	3.3	1.4	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	3.4	1.4	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	3.3	1.3	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	3.3	1.3	0.6	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	3.3	1.3	0.6	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	3.3	1.3	0.6	0	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	33.3	34.5	17.6	3.4	0	11.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	36.4	33.8	15.6	2.9	0	11.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	33.1	33.9	19	2.6	0	11.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	37	32.7	16.5	2.4	0	11.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.1	33.9	17.8	2.9	0	11.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	38.5	31.6	16	2.7	0	11.2
U2 - Efficacy additive alpha prior Beta(1, 1)	34.3	33.8	18.2	2.4	0	11.3
U3 - Efficacy additive alpha prior Beta(1, 1)	38.4	32.1	15.9	2.2	0	11.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.9	7.8	6.2	1.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13	7.8	6.2	1.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13	7.7	6.2	1.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13	7.7	6.3	1.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.1	7.6	6.3	1.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.1	7.5	6.3	1.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	13.1	7.6	6.3	1.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	13.1	7.5	6.3	1.6	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.1	1	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.1	1	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.1	1	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.1	1	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.2	1	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.1	1	0.6	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.1	1	0.6	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.1	1	0.6	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.1	31.8	34.2	5.3	0.1	5.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.4	32.5	30.5	4.9	0.1	5.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.4	31.6	36.1	5.2	0.1	5.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.6	32.4	32.1	4.2	0.1	5.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.8	30.4	36.8	4.3	0.1	5.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	28	30.9	31.7	3.7	0.1	5.6
U2 - Efficacy additive alpha prior Beta(1, 1)	22.8	29.8	36.7	5	0.1	5.6
U3 - Efficacy additive alpha prior Beta(1, 1)	27.6	29.8	32.9	4	0.1	5.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.4	8	8.7	2.7	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	8	8.8	2.6	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	7.9	8.8	2.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	7.9	8.8	2.7	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.5	7.7	8.8	2.7	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.5	7.7	8.8	2.7	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.5	7.7	8.8	2.7	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.5	7.7	8.8	2.7	0.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.2	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.2	1.5	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.2	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.2	1.5	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.2	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.2	1.5	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.2	1.4	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.2	1.5	0.8	0.1	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.2	36	2.3	0	0	39.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.1	36.8	1.4	0	0	39.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.9	36.4	2.3	0	0	39.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.5	35.8	1.8	0	0	39.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.7	36.3	2.6	0	0	39.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.9	35.7	2	0	0	39.4
U2 - Efficacy additive alpha prior Beta(1, 1)	21.2	36.9	2.4	0	0	39.5
U3 - Efficacy additive alpha prior Beta(1, 1)	22.7	35.8	2	0	0	39.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.5	7.5	2.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.6	7.5	2.4	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.6	7.4	2.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.6	7.5	2.5	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.7	7.4	2.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.6	7.4	2.5	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	14.6	7.4	2.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	14.7	7.4	2.5	0.3	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.7	0.5	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.6	0.5	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.7	0.5	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.7	0.5	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.7	0.5	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.7	0.5	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.7	0.5	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.7	0.5	0.1	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	67.4	27.7	1.9	0.1	0	2.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	69	26.6	1.4	0.1	0	2.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	69.3	25.8	1.9	0.1	0	2.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	71.9	23.6	1.4	0.1	0	3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	70.8	24.4	1.7	0.1	0	3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	72.7	23.1	1.1	0.1	0	3
U2 - Efficacy additive alpha prior Beta(1, 1)	70.7	24.7	1.5	0.1	0	3
U3 - Efficacy additive alpha prior Beta(1, 1)	73	23	0.9	0.1	0	3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.8	7	2.3	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.8	7.1	2.3	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20	6.9	2.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.9	6.9	2.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.1	6.7	2.3	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.1	6.7	2.3	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	20	6.8	2.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	20.1	6.8	2.3	0.2	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.1	0.6	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.1	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.1	0.6	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.1	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2	0.6	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.1	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2	0.6	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	2	0.6	0.1	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.3	21	18.3	35	6.6	0.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.6	24	14.7	31.4	4.5	0.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.2	18.7	18.5	36.5	6.3	0.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.1	22.2	14.9	31.9	5.1	0.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.6	17.6	18.8	36.9	6.3	0.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.1	20.5	15.2	32.1	5.3	0.8
U2 - Efficacy additive alpha prior Beta(1, 1)	19.4	17.6	19.7	36	6.5	0.8
U3 - Efficacy additive alpha prior Beta(1, 1)	25.8	20	15.9	31.5	6	0.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	5.5	6.7	8.2	4.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	5.5	6.7	8.1	4.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.5	5.4	6.7	8.2	4.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.5	5.3	6.8	8.3	4.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.5	5.3	6.7	8.2	4.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.5	5.3	6.8	8.1	4.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.5	5.4	6.8	8.2	4.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.4	5.3	6.7	8.3	4.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3	1.7	2.3	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3	1.7	2.3	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3	1.7	2.3	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3	1.7	2.3	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3	1.7	2.3	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3	1.7	2.3	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.1	1.7	2.3	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3	1.7	2.3	0.8	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.8	36.6	27.5	25.6	5.4	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.3	43.7	24.9	20.6	4.4	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.6	32.9	29.1	26.6	5.7	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.5	40.3	25.2	22.1	4.8	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.5	32.1	30.1	26.4	5.8	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.9	37.6	27.5	21.7	5.2	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	5.3	32.3	29.2	26.5	6.6	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	7.3	39	26.7	22.3	4.6	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	6	6.7	8.2	4.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	6.1	6.8	8.1	4.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.1	6	6.9	8.1	5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.1	6	6.9	8.2	4.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	5.9	7	8.2	4.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	5.8	6.9	8.1	5	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.1	5.9	6.9	8.2	5	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.1	5.9	6.9	8.2	5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.9	1.5	1.9	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.9	1.5	1.9	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.8	1.5	1.9	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.8	1.5	1.9	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.8	1.5	1.9	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.8	1.4	1.9	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.8	1.5	1.9	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.8	1.5	1.9	1.1	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9	17.1	32.7	32.4	8.2	0.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14	20	30.5	30.1	4.8	0.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.5	15.5	33.3	33.2	7.9	0.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14	19.7	31.1	29.5	5.1	0.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.2	14.4	35.6	31.2	9	0.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.3	18.7	33.5	26.2	5.7	0.6
U2 - Efficacy additive alpha prior Beta(1, 1)	9.3	14.5	36.2	30.7	8.7	0.6
U3 - Efficacy additive alpha prior Beta(1, 1)	15.1	19	32.9	26.9	5.5	0.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.9	4.7	7.8	8	5.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.9	4.7	7.8	8	5.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.9	4.6	8	7.8	5.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.9	4.6	7.9	7.8	5.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.9	4.5	8	7.8	5.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.9	4.5	8	7.8	5.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.9	4.4	8.1	7.8	5.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.9	4.5	8	7.6	5.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.9	2	2.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.9	2	2.8	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.9	2	2.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	2	2.8	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.9	2	2.8	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.9	2	2.7	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.8	2	2.7	1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.9	2	2.7	1.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.7	9	15.6	47.2	21.5	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.1	11	15.1	44.1	17.7	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.5	8.3	16.6	47	21.6	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.7	9.7	14.3	45.7	18.6	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6	8.5	17	45.2	23.3	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.8	9.8	15.8	44.6	19	0
U2 - Efficacy additive alpha prior Beta(1, 1)	5.8	8.2	17.4	46.2	22.4	0
U3 - Efficacy additive alpha prior Beta(1, 1)	10.8	9.9	14.6	45.4	19.3	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.7	5.7	9	8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.7	5.6	9	8.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.7	5.6	9	8.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.7	5.7	9	8.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.7	5.7	8.8	8.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.7	5.6	8.8	8.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.7	5.7	8.8	8.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.7	5.7	8.7	8.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.9	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.9	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.9	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.9	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.9	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.4	2.9	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.4	2.9	1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.4	2.9	1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.7	20.8	46.7	12.7	2.1	2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.3	21.3	42.1	11.8	1.4	2.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.5	19.9	48.1	12.2	2.3	2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.9	20.2	42	11.1	1.7	2.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.7	21.4	48	12.7	2.1	2.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.1	20.3	43	10.8	1.7	2.1
U2 - Efficacy additive alpha prior Beta(1, 1)	13.8	21.5	48.1	12.2	2.3	2.1
U3 - Efficacy additive alpha prior Beta(1, 1)	22.1	20.7	41.9	11.8	1.4	2.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.9	7.1	9.9	4.7	1.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6	7	10	4.6	1.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.1	6.9	10.1	4.6	2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6	7	10.1	4.7	1.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6	6.9	10.2	4.5	2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6	7.1	10.1	4.5	2	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.1	6.9	10.2	4.5	2	
U3 - Efficacy additive alpha prior Beta(1, 1)	6.1	7	10.1	4.5	2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.5	1.9	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.5	1.9	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.4	1.9	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.5	1.9	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.5	1.9	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.5	2	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.5	1.9	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.5	2	1.4	0.4	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10	47.4	31.4	9.8	1.3	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.9	51.4	25.9	9.2	0.5	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	46.7	32.7	9.9	1	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14	49.4	27.4	8.5	0.6	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.1	44.4	34.2	10.2	1	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.4	48.1	29.5	8.4	0.5	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	10.6	45	33.4	9.8	1.1	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	14.5	46.6	30.1	8.1	0.6	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	9.8	9.3	4.5	1.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.7	9.8	9.2	4.5	1.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	9.6	9.3	4.5	1.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	9.7	9.4	4.5	1.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.8	9.4	9.4	4.6	1.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.7	9.5	9.4	4.5	1.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.7	9.5	9.5	4.5	1.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.7	9.3	9.6	4.5	1.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.3	1.6	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.3	1.7	1.3	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.3	1.6	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.3	1.7	1.3	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.3	1.7	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.2	1.7	1.3	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.3	1.7	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.2	1.7	1.3	0.4	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.4	49.3	8.9	1.7	0.4	17.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.9	46.8	7.7	1.6	0.3	17.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	23	47.6	9.5	2.2	0.4	17.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.1	45.6	8.2	1.5	0.2	17.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.3	48.4	9.7	1.9	0.6	17.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.8	45.9	7.7	1.8	0.4	17.4
U2 - Efficacy additive alpha prior Beta(1, 1)	21.8	48.6	10	2	0.6	17
U3 - Efficacy additive alpha prior Beta(1, 1)	26.6	46.3	7.7	1.7	0.4	17.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.7	9	4.6	1.5	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.8	8.9	4.6	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.7	8.9	4.7	1.5	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.7	8.8	4.7	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.8	8.7	4.7	1.5	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.8	8.8	4.7	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.8	8.7	4.7	1.5	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.8	8.7	4.7	1.6	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.6	1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.6	1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.6	1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.6	1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.6	0.9	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.6	1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.6	1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.6	1	0.6	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.8	57.3	7.8	0.6	0.1	7.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.7	53.8	6.3	0.8	0	7.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.8	56.6	7.5	0.7	0	7.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.6	52.3	6.9	0.6	0	7.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.7	56.6	8.4	0.8	0	7.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.8	53	7	0.7	0	7.5
U2 - Efficacy additive alpha prior Beta(1, 1)	26.1	57.4	8.3	0.8	0	7.4
U3 - Efficacy additive alpha prior Beta(1, 1)	32.1	53.1	6.8	0.6	0	7.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.7	11.1	5.1	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.6	11.2	5.1	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.7	11.1	5	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.6	11.2	5.1	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.7	11.1	5.1	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.7	11	5.2	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.8	11	5.1	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.6	11.1	5.2	1.3	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.1	1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4.1	1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.2	1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.2	1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.2	1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.2	1.1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.2	1.1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.2	1	0.6	0.1	